

Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 1: Nutrition in Plants

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	1 — Nutrition in Plants
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic Paper-5 traps: forgetting respiration runs in the light too, so a net gas figure is read as one process; treating the compensation point as 'both stopped' rather than 'equal rates'; assuming soil nitrogen or air nitrogen is a same-day photosynthesis raw material; reading a positive starch test as proof of chlorophyll, or a negative one as proof food was never made when it was just locked behind an unsoftened waxy leaf; confusing xylem (water, up) with phloem (food, out) and confusing which controlled set-up isolates light versus chlorophyll versus carbon dioxide; calling a host-dependent parasite a saprotroph or a mutualist; crediting most new dry mass to soil rather than to carbon dioxide; and assuming an insectivorous or nodulated plant is not a true autotroph. Do not write on this paper — mark answers on your answer sheet.

1. In one bright hour a sealed-chamber leaf fixes carbon dioxide at 24 units per hour by photosynthesis while respiration releases it at 5 units per hour. Cloud then halves the light, and photosynthetic fixing falls in proportion to the light while respiration is unchanged. What does a meter reading only the NET carbon-dioxide change of the surrounding air now show per hour?

- (A) A net fall of 19 units, because respiration switches off whenever any light still reaches the leaf.
- (B) A net fall of 12 units, because only the photosynthetic uptake affects the air.
- (C) A net rise of 5 units, because halving the light makes respiration the only process that still runs.
- (D) A net fall of 7 units, because photosynthesis now fixes 12 while respiration still releases 5.

2. A leaf in a sealed chamber sits at its exact compensation point, where the meter shows zero net carbon-dioxide change. The experimenter slowly RAISES the light from this point. Before any other change, what does the meter begin to show, and why?

(A) Still zero, because the compensation point is fixed and does not shift when light changes.

(B) A net RISE in carbon dioxide, because brighter light makes the leaf respire faster to cope.

(C) A net FALL in carbon dioxide, because more light speeds photosynthesis above the steady respiration rate so uptake now exceeds release.

(D) A net RISE in carbon dioxide, because at high light the leaf stops photosynthesising to protect its chlorophyll.

3. A young tree adds 30 kg of dry mass in a year. A student's accounting: 18 kg from carbon dioxide, 6 kg traced to water supplying hydrogen, 1 kg of soil minerals, and the rest 'lifted as solid soil through the roots.' Using only this chapter's ideas, which single correction makes the accounting sound?

(A) Roots do not lift solid soil; the remaining mass also traces to carbon dioxide fixed in photosynthesis, not to whole soil carried up.

(B) The carbon-dioxide share is far too high; almost all dry mass actually comes from minerals dissolved in soil water.

(C) Water cannot supply any mass at all, so the 6 kg should be reassigned to oxygen the tree absorbed.

(D) The minerals figure is too low; soil minerals normally make up most of a tree's dry weight.

4. Four labels are claimed for a single fern that has chlorophyll-rich green fronds and grows in plain soil with no captured prey and no host: (i) autotroph, (ii) producer in a food chain, (iii) heterotroph, (iv) organism that needs raw materials from its surroundings. Exactly how many of these four are CORRECT for this fern?

(A) Two — only autotroph and producer, since a food-maker needs nothing from its surroundings.

(B) Four — all apply, because every green plant is also a heterotroph by night.

(C) One — only autotroph, since producers must be the largest organism present and a fern is small.

(D) Three — it is an autotroph, a producer, and it still needs raw materials; only the heterotroph label is wrong.

5. A scientist tags a single oxygen atom and finds it leaving a brightly lit leaf as part of the gas bubbling away. Tracing that atom BACKWARD to where it came from inside the leaf, the correct origin is:

(A) It came from the carbon dioxide the leaf took in, since that gas supplies both the carbon and the oxygen that bubble away.

(B) It came from a water molecule the plant used as a raw material; the oxygen released in photosynthesis is given off, not built into the food.

(C) It came from the glucose the leaf had just made, which breaks apart to release oxygen.

(D) It came from the soil minerals carried up to the leaf in the water stream.

6. A teacher writes: 'glucose + oxygen → (in sunlight, with chlorophyll) → carbon dioxide + water.' She says it contains TWO faults that, once fixed, give the correct photosynthesis word equation. The pair of corrections is:

(A) Reverse the arrow's sides so carbon dioxide + water are the raw materials and glucose + oxygen the products.

(B) Replace 'sunlight' with 'darkness' and replace 'chlorophyll' with 'soil minerals'.

(C) Replace 'glucose' on the left with 'starch' and 'oxygen' on the left with 'nitrogen'.

(D) Replace 'water' on the right with 'glucose' and keep everything on the left unchanged.

7. Two well-watered destarched plants on one sunny day differ only in their air: plant A breathes normal air, plant B breathes air enriched with extra carbon dioxide. Plant B's leaves test a deeper blue-black with iodine. A student concludes 'carbon dioxide must be the energy source for photosynthesis.' The correct reading of this result is:

(A) Extra carbon dioxide supplied more of a raw material, raising the food made; but the energy still comes from light, not from the gas.

(B) The student is right: more carbon dioxide means more energy, which is why more starch formed.

(C) The deeper colour proves only that iodine reacts more strongly when extra gas is dissolved in the leaf.

(D) Carbon dioxide replaced sunlight as the energy source, which is why plant B outperformed plant A.

8. A variegated leaf has a green ring around a white centre. Before a sunny day, a foil disc is pinned over PART of the green ring AND part of the white centre. After the day, the leaf is destarched-checked and given the iodine test. Which single region turns blue-black?

(A) The whole green ring, covered or not, because chlorophyll alone is enough to make starch.

(B) Both uncovered regions, green and white, because any tissue left in the light can make food.

(C) Only the green ring that was left uncovered, because starch needs both chlorophyll and light, met only there.

(D) Only the white centre that was left uncovered, because pale tissue stores the most starch.

9. A copper-beech leaf (deep red all summer) and a true albino white leaf-patch are both destarched, given the same sunny day, then tested with iodine. The red leaf turns blue-black; the white patch stays brown. The single correct reason for this difference is:

- (A) The red leaf still contains chlorophyll hidden under a red pigment and photosynthesises; the white patch genuinely lacks chlorophyll and cannot.
- (B) Red pigment is itself a form of chlorophyll, so both should have tested positive equally.
- (C) The white patch stores its starch invisibly, so it really did make food but iodine cannot show it.
- (D) The red leaf draws ready-made food from the trunk, which is why only it tests positive.

10. An experimenter wants ONE set-up that isolates chlorophyll as the needed factor while holding light, carbon dioxide and water identical across the compared tissues. Which design does this most cleanly?

- (A) Cover half of a uniformly green leaf with foil, give it a sunny day, then compare the two halves.
- (B) Keep one whole green plant in the dark and another in light, then compare a leaf from each.
- (C) Seal a green plant in a jar of normal air, light it, and measure the gas change inside.
- (D) Take a single variegated leaf and, after a sunny day, compare the starch test of its green parts with its white parts.

11. A leaf disc genuinely full of starch is dipped straight into iodine and barely changes colour. A second disc, equally full of starch, is boiled in water, then in alcohol, then dipped in iodine and turns deep blue-black. Which single statement correctly accounts for BOTH results?

- (A) The boiling steps manufactured the starch that the second disc shows, while the raw disc simply had none.
- (B) The boiling softened the tissue and stripped the waxy green coat so iodine could reach the starch; the raw disc's intact surface kept iodine out, hiding starch that was really there.
- (C) The alcohol added starch to the second disc, which is why only it reacted with iodine.
- (D) Iodine destroys starch in raw living tissue, so only the boiled, killed disc could keep its starch to react.

12. A child destarches a plant, then on a sunny day seals one leaf in carbon-dioxide-free air (a dish of potassium hydroxide inside) and leaves a second leaf in normal air. Only the normal-air leaf later turns blue-black. The child says 'the destarching step was pointless busywork.' The best response is:

- (A) The child is right; destarching changes nothing because potassium hydroxide alone decides the outcome.
- (B) Without destarching, leftover old starch could turn the carbon-dioxide-free leaf blue-black too, falsely suggesting it had made food — so destarching is what makes the result trustworthy.
- (C) Destarching mattered only because it re-greens the leaves so the iodine colour shows up.
- (D) Destarching mattered only because it opens the stomata so carbon dioxide can enter faster.

13. A pondweed in water releases 64, 16 and 4 oxygen bubbles per minute as a lamp sits at 10 cm, 20 cm and 40 cm, with water, temperature and the plant held constant. A student says 'the rate quarters each time the distance doubles because the plant is losing chlorophyll as the lamp retreats.' The best correction is:

- (A) The chlorophyll is intact and works again when the lamp returns; the rate falls because the light reaching the plant weakens as the lamp moves away, and rate tracks the available light.
- (B) The chlorophyll really is being destroyed, which is the only reason the bubble rate drops with distance.
- (C) The bubbles are carbon dioxide, and less of that gas forms as the lamp retreats.
- (D) The plant runs short of water as the lamp moves off, which slows the bubbling.

14. A leaf on a hot, dry, windy afternoon closes its stomata to conserve water. A student lists four effects: (i) carbon-dioxide intake drops, (ii) photosynthesis slows, (iii) sunlight can no longer reach the chlorophyll, (iv) water loss by transpiration falls. Exactly which of these actually follow from the stomata closing?

- (A) All four follow, because closed stomata seal the leaf against both gas and light.
- (B) Only (iii) and (iv) follow, since closing pores blocks light and water but never affects the carbon-dioxide supply.
- (C) Effects (i), (ii) and (iv) follow; (iii) does not, because light enters through the leaf surface, not the tiny pores.
- (D) Only (iv) follows, because transpiration is the single thing the stomata control.

15. Three leaves on one well-watered plant get a sunny day: leaf 1 greased thickly on its UPPER surface only, leaf 2 greased thickly on its LOWER surface only, leaf 3 untouched. Knowing most stomata sit on the lower surface, which leaf is expected to test the LEAST blue-black, and why?

- (A) Leaf 2, because greasing the lower surface clogs most of the stomata, cutting off the carbon dioxide needed to make starch.
- (B) Leaf 1, because the upper surface faces the sun and greasing it blocks the light that the chlorophyll needs.

(C) Leaf 3, because an untouched leaf has no protection and loses its starch fastest.

(D) All three equally, because grease anywhere simply stops iodine from reacting later.

16. A clear bag is tied over a living leaf on a sunny plant and droplets collect inside. An identical bag is tied over a freshly killed leaf on the same plant and far fewer droplets form. A student says 'the dead leaf makes less water inside the bag.' The correct account is:

(A) Correct as stated: the living leaf manufactures water from its sugar, which is why more droplets form.

(B) Neither leaf 'makes' water; the living leaf loses water vapour through its working stomata by transpiration, which the dead leaf's stomata can no longer drive.

(C) The droplets are simply rain that leaked into the living leaf's bag but not the dead one's.

(D) The living leaf's xylem squirts liquid water straight out of the leaf surface into the bag.

17. A scientist injects a harmless dye into the water at a stem's base and, the same hour, feeds a labelled sugar to a brightly lit upper leaf. Hours later the dye is high in the topmost leaves while the labelled sugar is found in the roots. Naming the tissues in the order (dye-carrier, sugar-carrier):

(A) Phloem carried the dye and xylem the sugar, since phloem moves water up and xylem moves food down.

(B) Xylem carried both, since one set of tubes moves everything up and down the plant.

(C) Stomata carried the dye and chlorophyll the sugar, since those are the plant's transport channels.

(D) Xylem carried the dye, since it moves water upward; phloem carried the sugar, since it moves food away from the leaves, here downward.

18. A ringed branch has a complete band of OUTER phloem stripped all the way round, with the inner xylem left whole; its leaves stay in full sun. A gardener predicts the swelling will appear BELOW the ring as sugars drain down to it. What actually happens, and why?

(A) The gardener is right: the swelling forms below the ring, where the descending sugars finally collect.

(B) No swelling forms anywhere, because cutting the phloem stops the leaves from making food at all.

(C) The swelling appears just ABOVE the ring, because leaf-made food moving down in the phloem is blocked at the cut and piles up on the leaf side.

(D) The swelling forms below the ring because the xylem pumps the leaf's sugars downward past the cut.

19. A student claims a leaf's broad, flat, thin shape with green cells packed near the upper surface evolved 'mainly to lose water fast and keep the plant cool.' Within this chapter, the best correction is:

- (A) These features mainly maximise sunlight capture for photosynthesis; the chapter explains leaf shape as an adaptation for food-making, not for cooling.
- (B) The student is right: the flat shape is chiefly a radiator built to shed water and heat.
- (C) The flat shape is mainly to store large amounts of water inside the thin leaf.
- (D) The packed green cells are mainly there to make the leaf heavy and resist the wind.

20. Four sealed transparent jars stand in bright light for a day, each with one organism: jar 1 a green moss, jar 2 a leafless *Cuscuta* vine with no host, jar 3 a bracket fungus, jar 4 a green alga. In how many jars is the carbon-dioxide level expected to have FALLEN by the end?

- (A) Three — the moss, the alga and the *Cuscuta*, since the *Cuscuta* photosynthesises using stored food.
- (B) All four, because every living thing absorbs carbon dioxide while it is in the light.
- (C) Two — the green moss and the green alga, because only chlorophyll-bearing organisms photosynthesise faster than they respire in the light.
- (D) One — only the green alga, since mosses are too small to lower the carbon dioxide measurably.

21. A leafless yellow *Cuscuta* thrives on a living shrub but withers on a dead, dry fence post; a bracket fungus flourishes on that same dead post and harms nothing alive. The classification fitting BOTH organisms is:

- (A) *Cuscuta* is a saprotroph that feeds on the dead post; the fungus is a parasite on the wood.
- (B) Both are parasites, since both lack chlorophyll and draw food from what they sit on.
- (C) *Cuscuta* is a parasite needing a living host; the fungus is a saprotroph feeding on dead, decaying matter.
- (D) *Cuscuta* is a symbiont helping the shrub; the fungus is an autotroph making food in its body.

22. A lichen is a fungus-and-alga partnership: the alga makes food, the fungus supplies water, minerals and shelter. A pea-*Rhizobium* nodule is also a partnership: the bacteria fix nitrogen, the plant supplies food and shelter. A student says 'both partnerships are parasitic because one side lives off the other.' The strongest single objection is:

- (A) Only the lichen is a partnership; the nodule is really the plant parasitising the bacteria.
- (B) Neither is a partnership at all, since a fungus and bacteria are not living organisms.
- (C) Both are parasitic, because in each case the food-maker is being robbed by the other.
- (D) In each partnership BOTH sides gain, so both are symbiosis, not parasitism, in which one partner is harmed.

23. An organism makes its OWN food yet is usually studied as the food-supplying member of a two-partner alliance. Among these, which fits that description, and what does its

partner provide?

- (A) The alga in a lichen makes the food; its fungal partner provides water, minerals and shelter.
- (B) Rhizobium in a nodule makes the food by photosynthesis; the legume provides the bacteria with light.
- (C) The fungus in a lichen makes the food; the alga provides it with shelter inside the rock.
- (D) Cuscuta makes its own food; its host plant provides it only with extra water.

24. A green pea with Rhizobium nodules, a green pitcher plant in a bog, and a leafless Cuscuta on a shrub all relate to nitrogen. Which single statement correctly says, for EACH, whether it makes its own food AND how it gets its nitrogen?

- (A) All three make their own food and all three fix nitrogen straight from the air through their leaves.
- (B) None of the three makes its own food; each relies entirely on another organism for both food and nitrogen.
- (C) Only Cuscuta makes its own food; the pea and pitcher plant are heterotrophs that cannot photosynthesise.
- (D) The pea and pitcher plant both photosynthesise; the pea gets nitrogen from its nodule bacteria, the pitcher plant from digested insects, while Cuscuta makes no food and takes nitrogen with everything else from its host.

25. A bog-dwelling pitcher plant with bright green leaves traps insects. A botanist removes its insect supply but gives it abundant light, water and carbon dioxide in nitrogen-poor soil. Over weeks the plant survives but grows pale and stunted. The best interpretation combining all facts is:

- (A) Cut off from insects, it must starve quickly, since the trapped insects were its main food source.
- (B) It keeps photosynthesising for energy-food, so it survives; but without insects its nitrogen-poor soil leaves it short of nitrogen, so it grows pale and stunted.
- (C) It should grow normally, because a green plant gets everything it needs from light, water and carbon dioxide alone.
- (D) It turns into a parasite on neighbouring roots to replace the lost insect food.

26. Soil in a sealed pot is sterilised to kill all nitrogen-fixing bacteria, then a bean is grown there in full light and water with no fertiliser. A student predicts the bean will die within a day for lack of carbon dioxide. What actually happens over weeks, and why?

- (A) It dies within a day, exactly as predicted, because sterilising the soil removes its carbon-dioxide supply.
- (B) It grows faster than normal, because removing soil bacteria leaves more nitrogen free for the plant.

(C) It cannot photosynthesise at all, because nitrogen is a raw material of the day's food-making.

(D) It survives but grows pale and stunted, because it still gets carbon dioxide from the air for daily photosynthesis but lacks the soil nitrogen needed for healthy growth.

27. Air is about 78% nitrogen gas, yet a tomato in nitrogen-rich air but nitrogen-poor soil stays pale and stunted. Two students disagree: one says 'air nitrogen is a chemically different element from the nitrogen plants need'; the other says 'most plants simply cannot take up nitrogen gas directly — it must be fixed into the soil first.' Who is right and why?

(A) The first is right: the nitrogen in air really is a different element from the nitrogen in soil.

(B) The second is right: it is the same element nitrogen, but the gas form is unusable until bacteria (or fertiliser) fix it into a soil form plants can absorb.

(C) Both are wrong, because tomato plants do not actually need nitrogen for healthy growth.

(D) Both are right, since air nitrogen is both a different element and impossible to absorb.

28. A farmer's field, cropped only with wheat for years, yields less each season as one nutrient drains. She wants to restore it without chemical fertiliser and considers four moves. Which works, and by what mechanism?

(A) Sow the wheat far more densely, so the thicker crop pulls extra nitrogen up from deep soil.

(B) Flood the field with extra water all summer, so the water carries nitrogen gas from the air into the soil.

(C) Grow a legume one season and plough it back, so *Rhizobium* in its root nodules fixes air nitrogen into the soil.

(D) Burn off the stubble after each harvest, so the ash returns the lost nitrogen to the soil.

29. In a food chain rice → grasshopper → frog → snake, every animal's energy traces back to one origin. A student insists 'the soil minerals the rice absorbs are the true energy source, since without soil the rice would not grow.' The best correction is:

(A) Soil minerals keep the rice healthy but are not the energy source; the energy originates as sunlight, captured by the rice in photosynthesis and passed along the chain.

(B) The student is right: soil minerals supply the energy that flows through the whole chain.

(C) The true energy source is the oxygen each animal breathes in as it moves along the chain.

(D) The true energy source is the bodies of the prey animals, which create new energy as they are eaten.

30. A pond holds green algae, tadpoles that eat the algae, and fish that eat the tadpoles. Persistent silt now shades the pond so almost no light reaches the algae. Tracing the chain of consequences over the following weeks gives:

- (A) The fish increase first, because shaded water becomes too dim for them to be hunted.
- (B) Shaded algae photosynthesise little and decline, so the tadpoles that feed on them run short and decline, and the fish that eat the tadpoles then decline too.
- (C) The tadpoles thrive, because algae shaded from light become softer and easier to digest.
- (D) Nothing changes, because algae make their food from the silt rather than from light.

31. A balanced sealed terrarium with a green plant thrives for weeks in good light, then is moved to a dark cupboard. A student predicts oxygen will build up because 'the resting plant releases stored oxygen at night.' What actually happens over two weeks, and why?

- (A) Oxygen builds up exactly as predicted, because a resting plant slowly lets out the oxygen it stored by day.
- (B) Nothing changes, because a plant in the dark pauses every life process and exchanges no gas.
- (C) Carbon dioxide falls, because in the dark the plant switches entirely to absorbing it for night-time food.
- (D) Oxygen falls and carbon dioxide rises, because in the dark photosynthesis stops while respiration keeps consuming oxygen and releasing carbon dioxide with nothing to recycle them.

32. Three destarched plants are given one bright sunny day, differing only as follows: plant 1 has both moist soil and normal air; plant 2 has moist soil but sits in carbon-dioxide-free air; plant 3 has normal air but bone-dry soil. Their leaves are then iodine-tested. Which plants test blue-black, and why?

- (A) All three, because bright light on its own is enough to make starch in any leaf.
- (B) Plants 1 and 2, because removing carbon dioxide does not stop food-making as long as the soil is moist.
- (C) Plants 1 and 3, because dry soil forces a leaf to make extra starch to survive.
- (D) Only plant 1, because starch needs light PLUS both raw materials — carbon dioxide and water — so removing either one (the gas from plant 2, the water from plant 3) stops it.

33. A leaf's net oxygen exchange is tracked over 24 hours: oxygen is released net from about 7 a.m. to 6 p.m. and taken in net otherwise, crossing zero at exactly two instants. At one of those instants the experimenter quickly DIMS the light. In the next moment the meter most likely shows:

- (A) A continued zero reading, because the zero-crossing is a fixed balance that dimming cannot disturb.
- (B) A net oxygen RELEASE, because dimming the light makes the leaf respire less and so release spare oxygen.
- (C) No reading at all, because dimming the light stops both photosynthesis and respiration together.
- (D) A net oxygen INTAKE, because dimming drops photosynthesis below the steady respiration rate, so oxygen used now exceeds oxygen made.

34. In the starch test, after the leaf is boiled in alcohol it comes out stiff, brittle and almost white. The next instruction is to dip it in warm water before adding the iodine. What is the purpose of this warm-water dip?

- (A) It adds back the chlorophyll the alcohol removed, turning the leaf green again.
- (B) It softens the brittle, decolourised leaf so the iodine can spread evenly over it and react with any starch present.
- (C) It washes the starch out of the leaf so that only the leaf's surface is left to be tested.
- (D) It kills bacteria that would otherwise eat the leaf's starch before the iodine could act.

35. A student writes four claims about the starch test; exactly ONE is correct. Which?

- (A) Boiling the leaf in water first softens it and breaks its cells so iodine and alcohol can act properly later.
- (B) A positive iodine test on any tissue proves that tissue contains chlorophyll.
- (C) Boiling in water is the step that dissolves out the green chlorophyll.
- (D) Skipping the destarching step makes no difference to whether the result can be trusted.

36. An experiment must show that LIGHT (not chlorophyll, not carbon dioxide) is the missing factor when a leaf fails to make starch. Which single set-up isolates light with the fewest other changes?

- (A) Take one uniformly green, well-aired, well-watered leaf, cover half with foil, give a sunny day, then starch-test both halves.
- (B) Compare the green and white parts of a variegated leaf after a sunny day.
- (C) Seal one leaf in carbon-dioxide-free air and another in normal air, both lit, then starch-test them.
- (D) Grow one plant in nitrogen-rich soil and another in nitrogen-poor soil, then starch-test a leaf from each.

37. A child says: 'A leaf that turns deep blue-black with iodine must be a green leaf, and one that stays brown must never have made food.' Using a potato tuber and an unsoftened raw leaf disc as counter-examples, the correct judgement is:

- (A) Only the first half is wrong; a brown result always means no food was ever made.
- (B) Both halves of the rule are wrong: a chlorophyll-free potato tests blue-black, and a raw leaf that really made starch can test brown if iodine cannot reach the starch through the unsoftened surface.
- (C) Only the second half is wrong; blue-black truly proves the tissue is green.
- (D) The rule is entirely correct; both counter-examples actually obey it.

38. A plant is given carbon dioxide, water, intact chlorophyll and gentle warmth, but kept in TOTAL darkness for a full day; a destarched leaf is then tested with iodine. A classmate predicts blue-black 'because warmth can stand in for sunlight as the energy source.' The correct result and reason are:

- (A) Blue-black forms, exactly as predicted, because gentle warmth supplies the energy in place of light.
- (B) Blue-black forms, because chlorophyll on its own can build starch in the dark once carbon dioxide is present.
- (C) No blue-black colour forms, because photosynthesis needs LIGHT as its energy source and warmth cannot replace it.
- (D) A red colour forms, because darkness changes the way iodine reacts with leaf tissue.

39. A student claims 'only leaves can ever photosynthesise, because food-making is a leaf's special job.' Considering a green cactus stem, a potato (which is an underground stem) and an ordinary root, the most accurate correction is:

- (A) The student is right: no part of a plant except a true leaf can ever photosynthesise.
- (B) Any part with chlorophyll that receives light can photosynthesise — so a green cactus stem does, while the buried potato stem and the root, lacking light and chlorophyll, do not; leaves are the main but not the only food-making site.
- (C) Underground parts photosynthesise fastest, because the soil shields their chlorophyll from being bleached by strong sun.
- (D) Roots photosynthesise using soil minerals in place of light, which is why roots do not need to be green.

40. Comparing how a mango tree, a cow and a mushroom obtain food, which single statement correctly places all THREE in their nutrition modes?

- (A) The mango tree is an autotroph making food from inorganic raw materials in light; the cow is a heterotroph eating ready-made plant food; the mushroom is a heterotroph (saprotroph) feeding on dead, decaying matter.
- (B) All three are heterotrophs, since none of them can build food from carbon dioxide and water.
- (C) The mango tree and the mushroom are autotrophs, since both grow without eating other animals; only the cow is a heterotroph.
- (D) The cow is the autotroph, since it turns the grass it eats into its own food using sunlight.

41. Imagine every green plant on Earth suddenly vanished, while all animals, fungi and bacteria remained for a time. According to this chapter's ideas, which consequence would follow most directly?

- (A) The air would gain oxygen and lose carbon dioxide, since only animals would be left to release oxygen.
- (B) Almost nothing would change, because animals and fungi can build their own food once the plants are gone.
- (C) Fungi would begin to photosynthesise in the plants' place, keeping the food chains supplied.
- (D) Food chains would collapse for want of producers, and the air's oxygen would no longer be renewed, because photosynthesis is what makes new food and releases oxygen.

42. A gardener finds a pale, leafless vine that withered on a metal trellis but flourishes on a living rose, and a soft growth on a rotting log that harms nothing alive. A friend labels them 'two saprotrophs.' The single best correction is:

- (A) Only the log-dweller is a saprotroph; the vine is a parasite, because it needs a LIVING host and dies on lifeless metal.
- (B) Both are saprotrophs, since neither has green leaves and both live off whatever they sit on.
- (C) Only the vine is a saprotroph; the log-dweller is a parasite harming the dead wood.
- (D) Both are autotrophs that have only briefly lost their green colour.

43. For a leaf actively making food in bright light, each substance is either taken IN as a raw material, given OUT as a product, or is the food made and kept. Which row classifies carbon dioxide, oxygen, water and glucose entirely correctly?

- (A) Carbon dioxide OUT, oxygen IN, water OUT, glucose IN.
- (B) Carbon dioxide IN, oxygen OUT, water IN, and glucose is the food made and kept in the leaf — neither a raw material taken in nor a gas given out.
- (C) All four are taken in as raw materials, since a leaf absorbs everything it needs from outside.
- (D) Carbon dioxide and oxygen are both given out, while water and glucose are both taken in.

44. Why is a green plant placed at the START of a food chain while a saprotrophic mushroom usually is NOT, even though both are alive and both respire?

- (A) The mushroom cannot take part in any food chain at all, so it can never appear.
- (B) The mushroom makes far more food than the plant but releases it too slowly to start a chain.

(C) The green plant makes brand-new food from sunlight, introducing fresh energy into the chain, whereas the saprotroph only breaks down food that already existed.

(D) The green plant begins the chain by eating the mushroom, which is why the mushroom comes later.

45. A student argues 'a carnivore like a snake eats only other animals, so it does not really depend on green plants at all.' Which single line of reasoning best shows the student is wrong?

(A) The snake is wrong because every snake also nibbles plants directly to survive.

(B) The animals the snake eats themselves trace their food back to plants, and the oxygen the snake breathes comes from plant photosynthesis, so the dependence runs through the chain.

(C) The snake is right; only plant-eating animals depend on green plants.

(D) The snake depends on plants only for the minerals it takes straight from the soil they grow in.

46. A student lists six organisms and calls them all autotrophs: a mango tree, a green pond alga, a leafy desert cactus stem, photosynthetic blue-green microbes, a Rhizobium bacterium, and a green pitcher plant. Exactly how many of the six are NOT autotrophs?

(A) Two — the Rhizobium and the pitcher plant, because trapping insects means a plant cannot be an autotroph.

(B) Three — the Rhizobium, the cactus and the pitcher plant, because none has ordinary green leaves.

(C) One — only the Rhizobium is not an autotroph; the rest, including the green pitcher plant and the green cactus stem, make their own food.

(D) None — all six are autotrophs, since each is involved with photosynthesis in some way.

47. A leaf blade is threaded with a fine, branching network of veins that reaches almost every cell. Within this chapter, the best account of what these veins do is:

(A) They stiffen the leaf so it can snap shut on insects for food, like a pitcher plant.

(B) They are tiny light-pipes that carry sunlight from the leaf edge inward to the shaded cells.

(C) They carry water up to all the food-making cells through the xylem and carry the food those cells make away through the phloem.

(D) They store the leaf's starch overnight so the green cells need not hold any themselves.

48. A gardener says 'the fertiliser I add to the soil is the plant's food, just as the meals I eat are mine.' Using this chapter, the most accurate correction is:

- (A) The gardener is right: fertiliser is exactly the plant's food, absorbed ready-made through the roots.
- (B) Fertiliser supplies minerals such as nitrogen that keep the plant healthy, but it is not the plant's food — a green plant makes its own food by photosynthesis and takes only minerals and water from the soil.
- (C) Fertiliser is the plant's energy source, fully replacing sunlight.
- (D) Fertiliser is broken down in the roots into ready-made glucose, which the plant then uses as its food.

49. On a variegated leaf with a green border and white centre, a thin strip running from the green border INTO the white centre is also covered by foil for the sunny day. After destarch-checking and the iodine test, the blue-black region is:

- (A) The whole green border plus the foil-covered strip, because chlorophyll alone makes starch.
- (B) Only the green border that was NOT under the foil, since starch needs chlorophyll (absent in the white) and light (blocked under the foil).
- (C) The whole leaf except the foil strip, because the white centre stores starch wherever light reaches it.
- (D) Only the foil-covered strip, because shading concentrates the leaf's starch under the foil.

50. A destarched plant has Leaf X enclosed with a dish of potassium hydroxide, Leaf Y greased thickly on its lower surface, and Leaf Z left untouched, all on the same sunny plant; then all three are iodine-tested. Which leaves stay BROWN, and what single cause links them?

- (A) Only Leaf X stays brown, because greasing a leaf cannot affect its food-making in any way.
- (B) Only Leaf Y stays brown, because potassium hydroxide actually supplies a leaf with extra carbon dioxide.
- (C) All three stay brown, because being on a destarched plant prevents any leaf from making starch that day.
- (D) Leaves X and Y stay brown; both were starved of carbon dioxide — X because potassium hydroxide absorbed it, Y because greased lower-surface stomata could not let it in.

51. A student summarises: 'Xylem carries food up to the leaves where it is used, and phloem carries water down to the roots.' How many of the two tissue-and-direction claims are correct?

- (A) One is correct: xylem really does carry food up, but the phloem direction is wrong.
- (B) One is correct: phloem really does carry water down, but the xylem claim is wrong.

(C) Both are correct, since the two tubes simply share the work of moving food and water.

(D) Neither is correct: xylem carries water UP, and phloem carries food made in the leaves OUT and down — so both substance and direction are swapped.

52. A leaf in still, humid shade loses almost no water vapour yet still tests blue-black for starch after a lit day; a leaf on a hot, dry, windy day loses water vapour fast. A student says 'low water loss proves the shaded leaf made no food.' Taken together, the observations show:

(A) The student is right: a leaf that loses little water cannot have made any food.

(B) Transpiration is required for photosynthesis, so the two must always rise and fall together.

(C) Transpiration and photosynthesis are separate processes; the shaded leaf lost little water yet still made starch, so low water loss does not mean no food.

(D) Water loss and food-making are two names for the same single process in the leaf.

53. A child reasons: 'When a seed sprouts in a dark drawer it grows a shoot, so plants must be able to photosynthesise in the dark; later it dies because the drawer runs out of air.'

Which single correction fixes BOTH halves of the reasoning?

(A) The child is right about the dark photosynthesis but wrong about the air; it actually dies of thirst.

(B) The seedling did photosynthesise in the dark using stored chlorophyll, and it dies only when the drawer's carbon dioxide is used up.

(C) The seedling grew on food stored in the seed, not by photosynthesis; it dies because, with no light, it cannot make new food once that store runs out, not from lack of air.

(D) The damp drawer fed the seedling ready-made food, and it dies when that outside food supply is exhausted.

54. A student says: 'Since autotrophs make their own food, a sealed green plant in bright light needs absolutely nothing from outside to keep growing.' The single best correction, using this chapter, is:

(A) Correct as stated: a brightly lit green plant is wholly self-sufficient and needs nothing external.

(B) It actually needs to absorb ready-made food from the soil, just as heterotrophs take in food.

(C) It still draws raw materials from its surroundings — carbon dioxide and water — plus light energy, to build that food; an autotroph is not independent of its surroundings.

(D) It needs to trap small insects each day to get the energy to start photosynthesis.

55. Considering both food and gases together, which single statement best captures how green plants and animals depend on each other?

- (A) Animals depend on plants for food, but plants depend on animals for nothing at all.
- (B) Plants depend on animals for their food, while animals depend on plants only for shade.
- (C) Neither depends on the other; each lives entirely on its own resources.
- (D) Animals get food and oxygen that trace back to plant photosynthesis, while plants take in the carbon dioxide that animals (and the plants' own respiration) release — a two-way dependence.

56. A scientist follows ONE carbon atom from a starch grain stored in a potato all the way back to where it entered the plant. Reading the path in BACKWARD order (from starch to entry point), the correct sequence is:

- (A) Starch in the potato ← food carried in the phloem ← glucose built in a lit leaf ← carbon dioxide entering through a stomatal pore.
- (B) Starch in the potato ← water carried up the xylem ← glucose built in the leaf ← carbon dioxide absorbed by the roots from the soil.
- (C) Starch in the potato ← food carried in the xylem ← glucose built in the leaf ← oxygen entering through the stomata.
- (D) Starch in the potato ← minerals carried in the phloem ← glucose built in the leaf ← carbon dioxide drawn from the soil water.

57. A teacher runs three set-ups on destarched leaves of one plant for a sunny day: set-up 1 a leaf in carbon-dioxide-free air, set-up 2 a leaf with half its area foil-covered, set-up 3 a variegated leaf left fully exposed. Matching each to the single factor it isolates as needed for photosynthesis:

- (A) Set-up 1 isolates carbon dioxide, set-up 2 isolates light, and set-up 3 isolates chlorophyll.
- (B) Set-up 1 isolates light, set-up 2 isolates chlorophyll, and set-up 3 isolates carbon dioxide.
- (C) All three set-ups isolate the same factor — light — just in different ways.
- (D) Set-up 1 isolates chlorophyll, set-up 2 isolates carbon dioxide, and set-up 3 isolates light.

58. Two destarched leaves on one lit plant are compared: leaf P sits in normal air, leaf Q in carbon-dioxide-free air made by potassium hydroxide; only leaf P tests blue-black. The teacher then asks which OTHER single change to leaf Q would have made it test blue-black too, without removing the potassium hydroxide. The answer is:

- (A) Shining a much brighter light on leaf Q, since enough light can make starch even without carbon dioxide.
- (B) Watering the plant more heavily, since extra water can replace the missing carbon dioxide in food-making.

(C) None of the usual changes — with its carbon dioxide still absorbed, leaf Q lacks a raw material, so neither more light, more water, nor more warmth can make it form starch.

(D) Warming leaf Q gently, since added warmth lets the leaf make its own carbon dioxide inside.

59. A sealed jar holds a green plant AND a small snail in bright light for a day; the gases roughly balance and both survive. The same jar is then kept in the DARK for a day. Compared with the lit day, the most likely change and reason is:

(A) Oxygen rises further, because in the dark the plant and snail both release the oxygen they stored by day.

(B) Oxygen falls and carbon dioxide rises, because in the dark the plant stops photosynthesising while both it and the snail keep respiring and using oxygen.

(C) Nothing changes, because the snail and plant simply rest and exchange no gas in the dark.

(D) Carbon dioxide falls, because in the dark the plant absorbs the snail's carbon dioxide faster than ever.

60. A plant is sealed in a clear chamber with plentiful carbon dioxide, water and warmth, but the light is dimmed week by week until it is nearly dark; the plant slows, weakens and finally dies. A student blames 'the plentiful carbon dioxide poisoning it.' The best correction is:

(A) The student is right: too much carbon dioxide gradually poisons and kills the plant.

(B) The plant died because the steady carbon dioxide used up all of its chlorophyll.

(C) The plant turned into a saprotroph and starved when the chamber walls offered no food.

(D) Carbon dioxide is a raw material, not a poison; the plant fails because too little light leaves it unable to photosynthesise enough food to sustain itself.

Solutions — Science (Class 7), Chapter 1:

Nutrition in Plants — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	B	21	C	31	D	41	D	51	D
2	C	12	B	22	D	32	D	42	A	52	C
3	A	13	A	23	A	33	D	43	B	53	C
4	D	14	C	24	D	34	B	44	C	54	C
5	B	15	A	25	B	35	A	45	B	55	D
6	A	16	B	26	D	36	A	46	C	56	A
7	A	17	D	27	B	37	B	47	C	57	A
8	C	18	C	28	C	38	C	48	B	58	C
9	A	19	A	29	A	39	B	49	B	59	B
10	D	20	C	30	B	40	A	50	D	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Halving the light halves photosynthesis: 24 → 12 units fixed. Respiration is unchanged at 5 units released. The air sees the net effect: 12 taken in minus 5 put back = 7 removed, a net fall of 7. The idea that respiration stops in light is the day-respiration misconception — it never switches off, so 19 is wrong. Counting only the 12 photosynthetic units ignores the 5 the leaf simultaneously returns to the same air. Halving the light does not silence photosynthesis altogether, so treating respiration as the only active process and predicting a rise of 5 is wrong.

2. (C) At the compensation point photosynthetic uptake equals respiratory release. Raising the light speeds photosynthesis while respiration stays roughly steady, so uptake now outpaces release and the air loses carbon dioxide — a net fall. The compensation point is not a fixed reading; it is just the balance for one particular light level, so 'still zero' is wrong. Brighter light does not drive respiration up to match; respiration is not controlled by light intensity that way. High light makes the leaf photosynthesise more, not shut down to 'protect chlorophyll' — that is invented behaviour.

- 3. (A)** Plant dry mass is mostly glucose, starch and related substances whose carbon comes from carbon dioxide fixed in the leaves; roots take up water and dissolved minerals only, never whole solid soil that 'hardens' into wood — so the 'solid soil through the roots' line is the error, and that mass belongs with the carbon-dioxide share. Saying minerals dominate is the food-from-soil misconception the chapter overturns. Water does supply the hydrogen built into glucose, so denying it any mass is wrong, and oxygen is a product released, not a building block absorbed. Minerals are needed for health but form only a small fraction of dry weight, so calling them most of it is wrong.
- 4. (D)** A green fern makes its own food, so it is an autotroph and the producer that begins a food chain — but it still draws carbon dioxide, water and light from its surroundings, so that label holds too. Being a food-maker, it is not a heterotroph, so that one is false: three of four are correct. The claim that an autotroph needs nothing is the independence misconception — raw materials are still required. Respiring at night does not make a plant a heterotroph; heterotrophy means taking ready-made food, which it does not. Producer status depends on making food from sunlight, not on body size, so the 'largest organism' rule is invented.
- 5. (B)** In photosynthesis water and carbon dioxide are the raw materials; oxygen is released as a product and traces to the water the plant used. The carbon dioxide's carbon goes into glucose, so attributing the escaping oxygen to that gas confuses raw material with the released product. Glucose is the food made and stored, not a source the leaf breaks apart to bubble oxygen away during food-making. Soil minerals are taken up for health but are not the source of the oxygen gas a leaf releases.
- 6. (A)** The written equation is photosynthesis run backwards: it lists the food and oxygen as raw materials and the gases as products. Photosynthesis takes in carbon dioxide and water and gives out glucose and oxygen, so swapping the two sides fixes both faults at once. Sunlight and chlorophyll are the correct conditions above the arrow, so changing them is wrong. Starch is a later storage form, not the immediate raw material, and nitrogen is not part of this equation. Leaving the left side unchanged keeps glucose and oxygen as raw materials, which is exactly the error.
- 7. (A)** Carbon dioxide is a raw material; supplying more of it lets a well-lit leaf build more glucose and store more starch, deepening the iodine colour. But the energy that drives photosynthesis is light — the gas is consumed, not burned for energy. The extra colour reflects more starch actually made, not iodine reacting differently to dissolved gas. Carbon dioxide cannot stand in for sunlight as the energy source; without light no starch forms however much gas is present.
- 8. (C)** Starch forms only where every requirement is met. The white centre lacks chlorophyll, so it never makes starch, covered or not; the foil-covered green is starved of light. The one region with chlorophyll AND light is the uncovered green ring, and only it turns blue-black. Chlorophyll alone cannot make starch in shade — light is also needed —

so the whole-ring answer is wrong. The uncovered white tissue still lacks chlorophyll and cannot photosynthesise, so 'both uncovered regions' is wrong. Pale, chlorophyll-free tissue makes no starch, so the white-centre answer is backwards.

9. (A) Photosynthesis needs chlorophyll. A red leaf only looks non-green because a red pigment masks chlorophyll that is still present, so it makes starch and tests positive; a white patch contains no chlorophyll, so it makes none and stays brown. Calling red pigment 'a kind of chlorophyll' is false — colour is not the same as the pigment. White tissue does not make and hide starch; with no chlorophyll there is no photosynthesis to make any. A red leaf makes its own food rather than drawing it ready-made from the trunk, so that explanation misstates how leaves feed.

10. (D) On one variegated leaf the green and white parts share the same light, air and water; only chlorophyll differs, so comparing their starch tests isolates chlorophyll. Foil over half a green leaf changes LIGHT between the halves, not chlorophyll, so it isolates light instead. Dark-versus-light whole plants again test light, not the pigment. Sealing a plant and measuring gas demonstrates gas exchange but does not single out chlorophyll as the variable. Only the variegated-leaf comparison cleanly isolates chlorophyll.

11. (B) Both discs had starch; the difference is access. Boiling in water softens and breaks the cells, and alcohol removes the waxy green chlorophyll, so iodine can penetrate and reveal the starch in the boiled disc. The raw disc's intact waxy surface blocks iodine, so its real starch goes undetected. The boiling does not create starch — the question states both discs already had it. Alcohol removes chlorophyll; it does not add food. Iodine detects starch by turning blue-black rather than destroying it, so the 'iodine destroys starch' idea is invented chemistry.

12. (B) Destarching empties the leaves of pre-existing starch, so any blue-black at the end must be new starch made during the experiment. Skip it and stored starch could give the carbon-dioxide-free leaf a false positive, wrecking the comparison — so the step is essential, not busywork. Potassium hydroxide removes carbon dioxide but does not by itself guarantee the result is interpretable without a clean starch baseline. Destarching empties starch; it does not 're-green' a leaf. Stomata handle carbon dioxide regardless of stored starch, so destarching does not speed gas entry.

13. (A) Only the lamp distance changed, and the rate falls in step with the weakening light, showing light intensity controls photosynthesis; chlorophyll is not consumed by dimming and works again when the lamp returns. So the loss-of-chlorophyll story is wrong. The bubbles are oxygen, a product released in light — carbon dioxide is taken in, not bubbled out. The plant sits in water throughout, so it cannot 'run short of water'; the limiting factor is the light reaching it.

14. (C) Stomata are the pores for gas exchange and water-vapour loss. Closing them cuts carbon-dioxide intake (i), which starves photosynthesis (ii), and reduces transpiration (iv).

But light does not enter through the stomata — it passes through the leaf surface regardless — so (iii) is false. Treating closed pores as a light seal is the mistake in the 'all four' and the '(iii) and (iv)' options. Stomata control gas exchange as well as water loss, so limiting their job to transpiration alone is too narrow.

15. (A) Carbon dioxide enters mainly through the lower-surface stomata; greasing the lower surface clogs them, starving leaf 2 of carbon dioxide so it makes the least starch. Grease on the upper surface blocks few stomata and does not stop light, which passes through the surface anyway, so leaf 1 is little affected. The untouched leaf 3 keeps all its stomata and makes the most starch, not the least. The boiling steps clean the leaf before testing, so grease does not simply 'stop iodine reacting' for all three; the real difference is how much starch each leaf made.

16. (B) The droplets are water vapour lost by transpiration through the stomata, then condensed in the bag. A living leaf's stomata transpire; a dead leaf's no longer do, so fewer droplets form — neither leaf manufactures water. Water is a raw material the plant uses, not something it makes from sugar. The sealed, identical bags rule out leaked rain as the difference. Xylem carries water inside the plant but does not squirt liquid out of the surface; loss is as vapour through stomata.

17. (D) Water (with the dye) rises from the roots through the xylem, so the dye appears high in the leaves; food made in the lit leaf (the labelled sugar) travels away through the phloem, here down to the roots. The role-swap option reverses both tissues' jobs. A plant has two distinct transport tissues, so 'xylem for both' is wrong. Stomata are gas-exchange pores and chlorophyll is a pigment, not transport channels, so that option misnames the structures.

18. (C) Phloem carries food made in the leaves downward and outward; the ring breaks that path, so food coming down from the leaves cannot pass and accumulates on the leaf side — just ABOVE the ring. Below the ring is cut off from the descending food, so no pile-up there, making the gardener's prediction wrong. The leaves still get water through the intact xylem and keep photosynthesising, so they do not stop making food. The xylem carries water upward, not the leaf's sugars downward, so it cannot deliver sugar below the cut.

19. (A) A broad, flat, thin leaf spreads a large area to the sun, and chlorophyll-rich cells near the lit upper surface catch that light — the chapter presents this as an adaptation for capturing sunlight to make food. Water loss does occur through stomata, but the shape's purpose in this chapter is food-making, not cooling. A thin leaf is not a water store — fleshy organs store water. The shape makes a leaf catch wind, not resist it, so 'heavy to resist wind' is wrong.

20. (C) Carbon dioxide falls only where photosynthesis outpaces respiration. The green moss and green alga both have chlorophyll, so in bright light they pull more carbon

dioxide out than they release — two jars. *Cuscuta* lacks chlorophyll and, with no host, only respire, so it raises carbon dioxide; counting it as a photosynthesiser is wrong. A bracket fungus is a heterotroph that only respire, so it also raises carbon dioxide — 'all four' is wrong. The moss has chlorophyll and does photosynthesise; its size does not stop it lowering carbon dioxide, so 'only the alga' undercounts.

21. (C) *Cuscuta* thrives only on a LIVING host and dies on dead wood — the mark of a parasite. The fungus thrives on DEAD wood and harms nothing alive — the mark of a saprotroph. Calling *Cuscuta* a saprotroph contradicts its death on the dead post; it cannot feed on dead matter. Lumping both as parasites ignores that the fungus needs no living host. *Cuscuta* harms its host rather than helping it, so it is no symbiont, and the fungus has no chlorophyll, so it is no autotroph.

22. (D) Parasitism harms one partner; in both the lichen and the nodule each side benefits — the alga and bacteria gain shelter and supplies, the fungus and plant gain food or usable nitrogen — so both are symbiosis. The nodule is a genuine mutual partnership, not the plant parasitising the bacteria. Fungi and bacteria are living organisms, so denying that either is a partnership is false. Neither food-maker is 'robbed': each gives food and receives something it needs in return.

23. (A) The lichen's alga has chlorophyll and makes the partnership's food, while the fungus supplies water, minerals and shelter — so the alga fits 'makes its own food in an alliance'. *Rhizobium* does not photosynthesise; it fixes nitrogen and takes food from the legume, so it is not the food-maker. In a lichen it is the alga, not the fungus, that makes food. *Cuscuta* lacks chlorophyll and makes no food at all; it steals everything from its host.

24. (D) The pea and the pitcher plant are green and photosynthesise; the pea gets usable nitrogen from *Rhizobium* in its nodules, the pitcher plant gets nitrogen by digesting trapped insects. *Cuscuta* lacks chlorophyll, makes no food, and draws nitrogen with all its nutrition from a living host. Ordinary plants cannot fix air nitrogen through their leaves, so the 'all fix it through leaves' claim is false. The pea and pitcher plant clearly make their own food, so 'none makes its own food' is wrong. *Cuscuta* is the one that cannot photosynthesise, so calling it the sole food-maker inverts the facts.

25. (B) The green leaves photosynthesise, supplying energy-rich food, so the plant survives without insects — it does not starve, because insects were never its main food, only its nitrogen source. Losing that nitrogen source in poor soil leaves it short of nitrogen, hence the pale, stunted growth. 'Grows normally' ignores that nitrogen, though not a same-day photosynthesis raw material, is still needed for healthy green growth. The plant does not become a parasite on neighbours; it remains a green food-maker simply lacking nitrogen.

26. (D) Carbon dioxide comes from the AIR through the stomata, not from soil bacteria, so the bean keeps photosynthesising and does not die for lack of it — the day-prediction is

wrong. What it loses is usable soil nitrogen, needed for healthy green growth, so it grows pale and stunted. Killing nitrogen-fixing bacteria removes a nitrogen SOURCE, it does not free extra nitrogen, so faster growth is wrong. Nitrogen is needed for growth and colour but is not a same-day photosynthesis raw material, so 'cannot photosynthesise at all' overstates its role.

27. (B) It is one element, nitrogen, but most plants cannot use the gaseous form; nitrogen-fixing bacteria (or fertiliser) must convert it into a usable soil form before roots can absorb it — so the second student is right. The 'different element' claim is false; the element is the same, just in an unusable gaseous state. Tomatoes, like all plants, need nitrogen for healthy green growth, so denying the need is wrong. Since the 'different element' half is false, 'both are right' cannot hold.

28. (C) Continuous wheat drains soil nitrogen. Growing a legume, whose *Rhizobium* nodules fix atmospheric nitrogen, and ploughing it back restores nitrogen naturally — the chapter's crop-rotation remedy. Denser wheat drains nitrogen faster, not restores it. Water does not carry nitrogen gas into the soil in a usable form; fixing requires bacteria, not flooding. Burning stubble does not replace nitrogen — much is lost to the air as the stubble burns — so ash is not a reliable nitrogen return.

29. (A) Green plants capture sunlight's energy as food in photosynthesis; that captured energy passes up the chain as one organism eats another, so the origin is sunlight, not minerals. Minerals support the rice's health but carry no energy into the chain. Oxygen is used to RELEASE energy from food during respiration, but it is not itself the energy source. Prey animals merely transfer energy that originally came from the plant; eating does not create new energy.

30. (B) Light is the algae's energy source; persistent shade starves them, so the producers decline first. Their consumers, the tadpoles, then run short of food and decline, and the fish above them follow — the impact travels up from the base. The fish do not increase; losing the chain's base harms the whole chain. Less light means less algae, not softer, more nutritious algae for the tadpoles. Algae are autotrophs needing light, not silt particles, so 'food from the silt' is false.

31. (D) In the light the terrarium balanced because photosynthesis and respiration recycled each other's gases. In the dark photosynthesis stops but respiration continues, taking in oxygen and giving out carbon dioxide with nothing to replace or remove them, so oxygen falls and carbon dioxide rises. A plant does not store and 'let out' oxygen at night — respiration uses oxygen. Respiration never pauses, so 'nothing changes' is wrong. Photosynthesis needs light, so no 'night-time food-making' absorbs carbon dioxide in the dark.

32. (D) Photosynthesis needs light AND both raw materials — carbon dioxide and water. Plant 1 has everything and makes starch; plant 2 is starved of carbon dioxide (the gas raw

material) and plant 3 of water (the other raw material), so neither makes starch. Only plant 1 tests blue-black. Light alone is not enough — the raw materials must also be present — so 'all three' is wrong. Removing carbon dioxide DOES stop food-making, so 'plants 1 and 2' is wrong. Dry soil starves the leaf of a raw material rather than driving it to make extra starch, so 'plants 1 and 3' is wrong.

33. (D) A zero-crossing is the instant photosynthetic oxygen output equals respiratory oxygen use. Dimming the light drops photosynthesis below respiration, so oxygen used now exceeds oxygen made and the meter shows a net intake. The crossing is only a balance for that light level, not a fixed point, so dimming does disturb it. Respiration is not slowed by dimming light, so it does not 'release spare oxygen'. Respiration never stops with the light, so both processes do not halt together.

34. (B) Boiling in alcohol removes the green chlorophyll but leaves the leaf stiff, brittle and bleached; dipping it in warm water softens it again so iodine can spread evenly and reach any starch. Alcohol does not put chlorophyll back, and iodine is added after, not before, decolourising — so 'adds back the chlorophyll' is wrong. The dip does not wash starch out of the tissue; the starch is inside the cells and stays to react, so 'washes the starch out' is wrong. The warm-water step is about softening the leaf, not killing bacteria, so that explanation is invented.

35. (A) Boiling in water softens the leaf and breaks open its cells so the later alcohol and iodine steps can reach the tissue — this is the correct claim. A positive iodine test shows starch, which can occur in chlorophyll-free tissue such as a potato tuber, so it does not prove chlorophyll. Removing chlorophyll is the ALCOHOL step's job, not the water boil — a classic mix-up. Skipping destarching lets leftover starch give false positives, so it certainly does affect whether the result can be trusted.

36. (A) On one uniformly green leaf with the same air and water throughout, foil over half changes only the LIGHT between the lit and shaded halves, so comparing them isolates light. A variegated leaf differs in chlorophyll between its parts, so it isolates chlorophyll, not light. The carbon-dioxide-free versus normal-air pair isolates carbon dioxide. The nitrogen-soil comparison tests a growth nutrient, not the same-day light requirement. Only the half-foil set-up cleanly isolates light.

37. (B) A potato tuber has no chlorophyll yet tests deep blue-black because it stores starch, so a positive test does NOT prove the tissue is green — the first half fails. A raw leaf disc full of starch can test brown if its waxy, unsoftened surface keeps iodine out, so a brown result does NOT prove no food was made — the second half fails too. Hence both halves are wrong, and neither counter-example obeys the rule.

38. (C) Photosynthesis requires light as its energy source. With everything present but light, no food is made, so the destarched leaf stays starch-free and tests brown — warmth cannot stand in for light, so the classmate's prediction fails. Chlorophyll captures light

energy but cannot make starch in the dark on its own. Darkness does not change iodine chemistry; iodine turns blue-black only where starch is present, and here there is none.

39. (B) Photosynthesis happens wherever chlorophyll and light meet, so a green cactus stem makes food, while a buried potato (an underground stem) and an ordinary root get no light and have no chlorophyll, so they do not — leaves are the main site, not the only one. The claim that only true leaves can ever photosynthesise ignores green stems, so it is wrong. Underground parts are in darkness and cannot photosynthesise at all, let alone fastest, so the 'soil shields chlorophyll' option is false. Roots do not photosynthesise on minerals in place of light; minerals are not an energy source, so that option is invented.

40. (A) The mango tree builds food from carbon dioxide and water using light — an autotroph. The cow cannot make food and eats ready-made plant material — a heterotroph. The mushroom has no chlorophyll and feeds on dead, decaying matter — a heterotroph of the saprotroph kind. The tree clearly does build food from inorganic materials, so 'all three heterotrophs' is wrong. The mushroom makes no food, so it is not an autotroph alongside the tree. A cow digests grass; it does not photosynthesise, so it is not the autotroph.

41. (D) Green plants are the producers that make new food and release oxygen, so losing them would cut off the base of every food chain and stop the renewal of the air's oxygen — the most direct consequences. Without plants making oxygen, the air would not gain oxygen; animals only use it up, so 'air gains oxygen' is backwards. Animals and fungi cannot build their own food, so they would not carry on unchanged. Fungi have no chlorophyll and cannot photosynthesise, so they could not take the plants' place. Only the collapse-of-producers answer follows.

42. (A) The log-dweller thrives on dead, decaying wood and harms nothing alive — a saprotroph. The vine dies on lifeless metal and flourishes only on a living host — the mark of a parasite, not a saprotroph, since a saprotroph could feed on dead matter. So 'two saprotrophs' is wrong: only one fits. Dead wood cannot be 'harmed', so the log-dweller is no parasite. Both lack chlorophyll and make no food, so neither is an autotroph that 'lost its colour'.

43. (B) In photosynthesis the leaf takes IN carbon dioxide and water as raw materials and gives OUT oxygen as a product; glucose is the food it makes and keeps, not a gas given off nor a raw material absorbed. So the correct row is carbon dioxide IN, oxygen OUT, water IN, glucose kept. Swapping carbon dioxide and oxygen (option A) reverses which gas is taken in and which given out. A leaf does not absorb everything — oxygen and glucose are not raw materials taken in — so 'all four in' is wrong. Carbon dioxide is taken in (not given out) and glucose is made (not taken in), so the last row is wrong.

44. (C) Producers begin food chains because they capture sunlight and create new food, adding fresh energy. A saprotroph makes no new food; it only decomposes matter that

already held energy, so it is not a starting producer. Saprotrophs do feature in food webs as decomposers, so 'cannot take part at all' is too strong. A mushroom has no chlorophyll and makes no food, so 'makes far more food' is false. Green plants make their own food and do not eat mushrooms to begin a chain.

45. (B) Even a pure carnivore depends on plants indirectly: its prey (and its prey's prey) ultimately ate plants, so the energy in every meal traces to plant food, and the oxygen the snake breathes is released by plant photosynthesis — the dependence runs through the whole chain. A snake does not need to eat plants directly for this to hold. The claim that only herbivores depend on plants ignores this indirect chain. Snakes get no minerals 'straight from the soil'; the key dependence is plant-made food and plant-released oxygen.

46. (C) Rhizobium fixes nitrogen and takes food from the legume; it does not photosynthesise, so it is the one non-autotroph. The green pitcher plant photosynthesises and only uses insects for nitrogen, so it remains an autotroph. The green cactus stem photosynthesises in place of leaves, so it is an autotroph. The mango tree, green alga and photosynthetic microbes all make their own food. So exactly one is not an autotroph. Insect-trapping does not strip the pitcher plant of autotrophy, and lacking ordinary leaves does not stop the cactus stem from photosynthesising — so two and three overcount. Rhizobium does not photosynthesise, so 'all six' is wrong.

47. (C) A leaf's veins are its transport tissues: the xylem carries water up to all the food-making cells, and the phloem carries the food those cells make away to the rest of the plant. Veins do not snap shut on insects — that is a special trap, not what ordinary veins do. Veins are tubes for water and food, not light-pipes; light reaches the cells through the leaf surface, so the 'carry sunlight inward' option is invented. Veins transport food rather than storing the leaf's starch overnight, so that option is wrong.

48. (B) A green plant makes its own food by photosynthesis; from the soil it takes only water and minerals (like nitrogen), which keep it healthy but are not its food. So fertiliser is a mineral supply, not food — the gardener's analogy is the food-from-soil misconception the chapter overturns. Fertiliser is not absorbed ready-made as food, so 'exactly the plant's food' is wrong. The plant's energy comes from sunlight, not fertiliser, so 'replacing sunlight' is wrong. Roots do not turn fertiliser into glucose; glucose is made in the leaves by photosynthesis, so that option is invented.

49. (B) Starch forms only where chlorophyll AND light coincide. The white centre has no chlorophyll, so it never makes starch; the foil blocks light wherever it lies, including over part of the green border. The only region with both chlorophyll and light is the uncovered green border, so only it turns blue-black. Chlorophyll alone cannot make starch in the foil's shade, so including the covered strip is wrong. The white centre cannot photosynthesise even when lit, so 'whole leaf except the strip' is wrong. Shading does not concentrate starch; it prevents it, so the foil strip stays brown.

50. (D) Both interventions cut off carbon dioxide. Potassium hydroxide absorbs it from Leaf X's air; grease clogs the lower-surface stomata through which carbon dioxide enters Leaf Y. Starved of this raw material, neither makes starch, so both stay brown, while the untouched Leaf Z turns blue-black. Greasing the stomata-rich lower surface does affect food-making, so 'grease cannot affect it' is wrong. Potassium hydroxide **ABSORBS** carbon dioxide rather than supplying it, so the claim about Leaf Y is inverted. Destarching only empties old starch; a lit, well-supplied leaf (Z) still makes new starch, so 'all three brown' is wrong.

51. (D) Xylem carries water (and dissolved minerals) UP from the roots; phloem carries the **FOOD** made in the leaves outward and downward. The student swapped both the substances and the directions, so **NEITHER** claim is correct. Xylem carries water, not food, so the 'xylem carries food up' half is wrong. Phloem carries food, not water, so the 'phloem carries water down' half is wrong. The two tubes have distinct jobs and do not simply share the work, so 'both correct' is false.

52. (C) The shaded leaf transpired little but still tested positive for starch, so it clearly made food despite low water loss — disproving the student. The two are distinct processes: transpiration is water vapour leaving the stomata, photosynthesis is food being built, and one can be low while the other proceeds. Since the shaded leaf made food with low transpiration, transpiration is not required for photosynthesis and they need not rise and fall together. They are not one process under two names — the observation of low loss with successful food-making separates them.

53. (C) A seed carries a food store that fuels early growth even without light, so the shoot does **NOT** come from dark photosynthesis. Photosynthesis needs light, so in the dark the seedling cannot make more food once the seed's store is used up, and it dies — not from lack of air. So both halves of the child's reasoning are wrong. The death is from exhausted stored food, not thirst. Chlorophyll cannot make starch in the dark, so 'dark photosynthesis' is impossible. A damp drawer supplies moisture, not ready-made food, so it did not feed the seedling.

54. (C) Autotrophs make their own food but not from nothing: they need carbon dioxide and water as raw materials and light as the energy source, all from their surroundings — so 'needs nothing external' is the independence misconception. They do not take ready-made food from the soil; that is the food-from-soil misconception the chapter overturns. Ordinary green plants do not trap insects for energy; they photosynthesise using light. Only the raw-materials correction is right.

55. (D) The dependence runs both ways: animals get their food (directly or through a food chain) and the oxygen they breathe from plant photosynthesis, while plants use the carbon dioxide released by animals' respiration (and their own). Saying plants depend on animals 'for nothing' misses the carbon dioxide return, so that is incomplete. Plants make their own food rather than depending on animals for it, so 'plants depend on animals for food' is

wrong. The two are clearly linked through food and gases, so 'neither depends on the other' is false.

56. (A) Tracing backward: stored starch came from glucose carried as food in the PHLOEM, which was built in a lit leaf from carbon dioxide that entered through a stomatal pore. Water travels in the xylem, not food, and carbon dioxide enters the leaf through stomata, not the roots — so the xylem/roots path is wrong. Food moves in the phloem, not the xylem, and the carbon comes from carbon dioxide, not oxygen, which is a product released. Minerals are not the food carried in phloem, and carbon dioxide is drawn from the air through stomata, not from soil water.

57. (A) Removing carbon dioxide (set-up 1) tests the carbon-dioxide requirement; foil over half a uniformly green leaf (set-up 2) varies only light between the halves, isolating light; a variegated leaf (set-up 3) varies only chlorophyll between green and white parts, isolating chlorophyll. The other matchings mislabel these: removing a gas does not isolate light, foil does not isolate chlorophyll, and a variegated leaf does not isolate carbon dioxide. The three do not all isolate light — each targets a different factor, which only the correct matching gets right.

58. (C) Carbon dioxide is an essential raw material; while the potassium hydroxide keeps absorbing it, leaf Q cannot build glucose no matter how much light, water or warmth it gets, so it stays brown — none of those changes helps. Light is the energy source but cannot substitute for the missing carbon-source gas. Water supplies hydrogen, not the carbon, so it cannot replace carbon dioxide. A leaf does not manufacture its own carbon dioxide from warmth; the gas must come from the air, which the potassium hydroxide is stripping away.

59. (B) In light, the plant's photosynthesis recycled the carbon dioxide and supplied oxygen for both organisms. In the dark photosynthesis stops, yet the plant and the snail both respire — using oxygen and releasing carbon dioxide — so oxygen falls and carbon dioxide rises. Neither organism 'stores and releases' oxygen at night; respiration consumes oxygen. Respiration never pauses, so 'nothing changes' is wrong. Without light there is no photosynthesis to absorb carbon dioxide, so it rises rather than falls.

60. (D) Light is the energy source for photosynthesis; as it fades the plant makes less and less food despite ample carbon dioxide, water and warmth, so it weakens and dies — light is the limiting factor. Carbon dioxide is a raw material the plant uses, not a poison, so the student's diagnosis is wrong. Keeping carbon dioxide topped up does not consume chlorophyll; the two are unrelated. A green plant does not turn into a saprotroph feeding on chamber walls — that is not a real mode by which a green plant lives.